

NIT RAIPUR INTERNAL HACKATHON 2026

- Problem Statement ID- **PS 53**
- Problem Statement Title- **Energy Theft and Abnormal Consumption Detection**
- Theme- **Renewable/Sustainable Energy**
- PS Category- **Software**
- Team ID- **SIH26-S073**
- Team Name- **Turing Sparks**



PROBLEM

EXISTING SOLUTION

PROPOSED SOLUTION

1. A team can visit only about 20 houses a day, but current tools flag thousands of meters with no order.

2. A broken meter, a meter with no signal, an empty house and a family using more all look unusual.

3. The only known thieves are the ones already caught, so the software learns to find that same kind again.

4. Acting on theft is a legal process, so an officer needs proof he can show and defend, not a score.

Software scores each house on its own and gives a long list of suspects. It never says who to visit first.

Most systems only answer yes or no to theft, so all these very different situations get the same alarm.

Systems are trained only on caught cases and assume everyone else is honest, which is not true.

The software gives a number with no explanation and no way to confirm a suspicion before acting.

We check whether units are actually missing on that street, then rank cases by how much money can be recovered.

We work out what is actually wrong and send each case to the right team: inspection, repair, network, or nobody.

We teach the software what normal looks like, treat unchecked houses as unknown, and learn from every inspection result.

Each case gets a one-page sheet in plain language, plus a switch test that proves a meter is being bypassed.

Uniqueness & Innovation

1



PHYSICS CROSS-CHECK

Units dispatched – units billed – computed i²R loss = unaccounted energy. Published models are meter-centric and ignore network topology.

2



NAMES THE CAUSE

Separates theft, rigged meter, failing meter, comms loss, vacancy and real load growth — routed to four teams, not one suspect list.

3



LEARNS WITHOUT LABELS

Autoencoder + PU-learning treat unchecked consumers as unknown, not innocent; every inspection outcome returns as a new label.

4



PHYSICAL CONFIRMATION

An operator-authorized relay test checks whether supply truly stops at the meter — proof of bypass, never automatic disconnection.

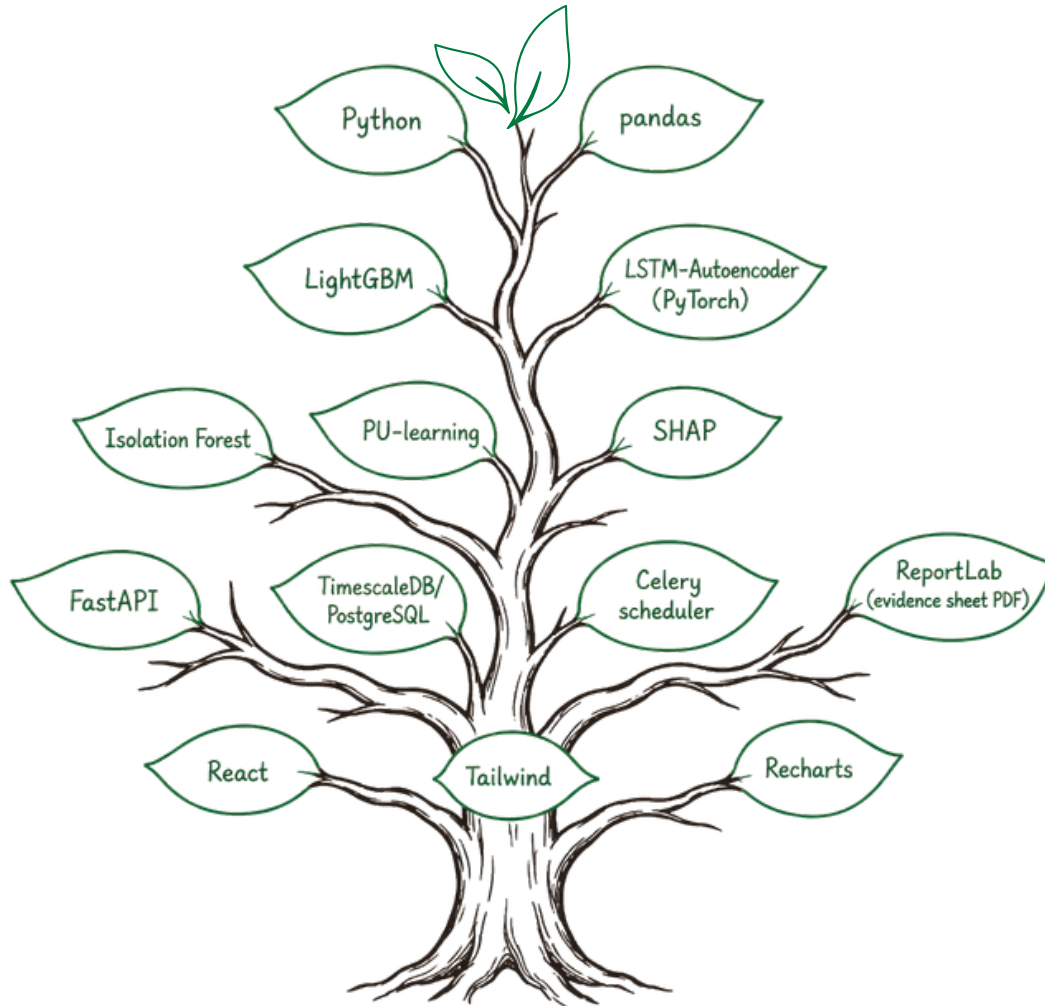
5



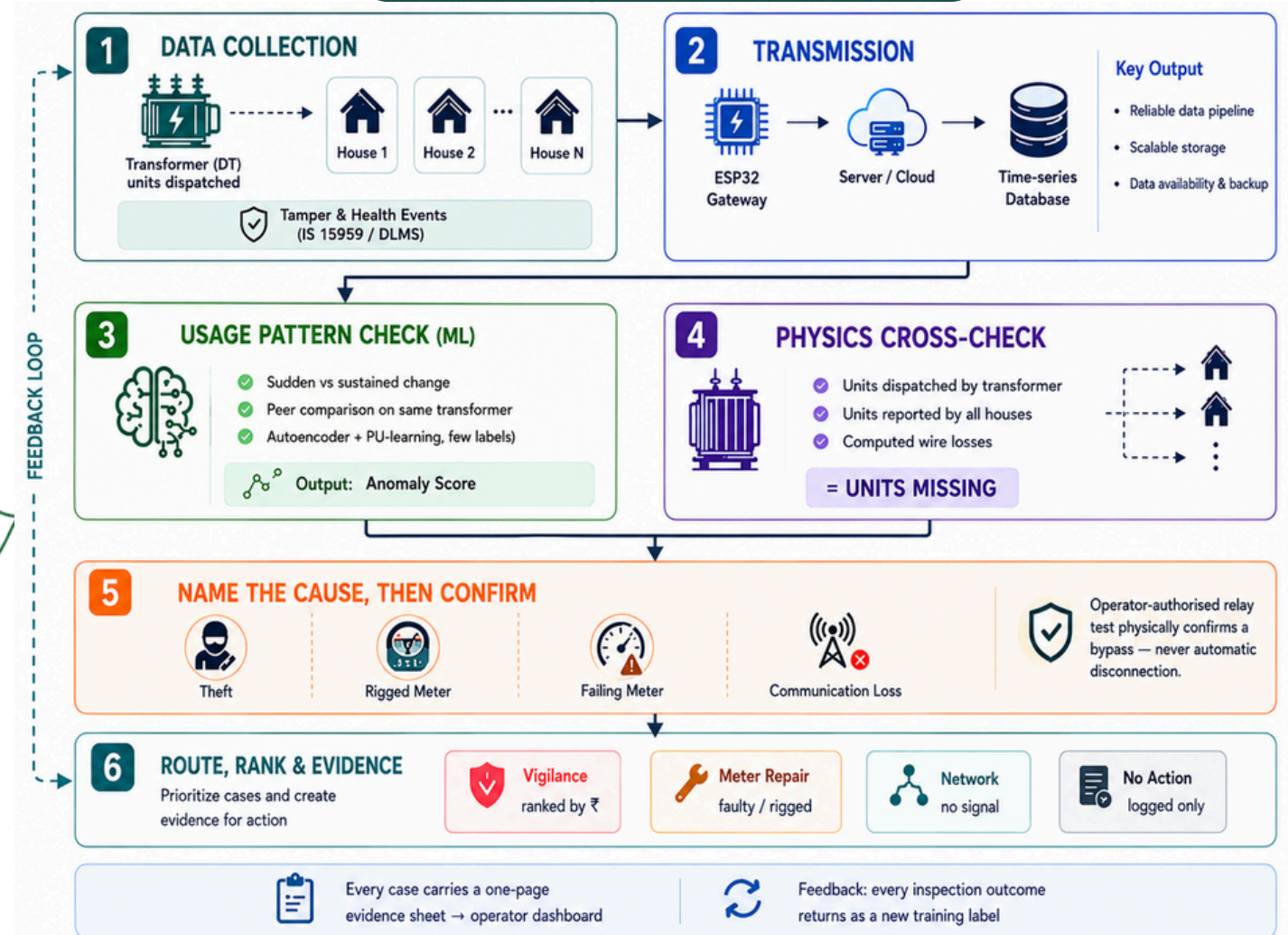
DECISION-GRADE OUTPUT

Ranked by ₹ recoverable net of inspection cost, with a SHAP-backed evidence sheet for ₹126/₹135 proceedings.

Tools & Technologies



Implementation



FEASIBILITY



Technical Feasibility

- ✓ SGCC dataset: 42,372 consumers × 1,036 days
- ✓ 3 model types, all open-source libraries
- ✓ 6 simulated theft patterns for training



Economic Feasibility

- ✓ ₹0 licensing — fully open-source stack
- ✓ 59.7 M smart meters already installed nationally
- ✓ ₹1.3 trillion of smart metering already approved



Operational Feasibility

- ✓ ~20 inspections/day — the real constraint we optimise
- ✓ 4 work queues instead of 1 suspect list
- ✓ 1-page evidence sheet per case

CHALLENGES



No labelled theft data

Risk: Model trains on the wrong patterns and fails on real data



False alarms waste time

Risk: Utility stops trusting the list; honest families get harassed



Score is not proof

Risk: Field officer cannot defend the case if questioned



Consumer privacy

Risk: Detailed usage data reveals household habits

STRATEGIES

Use public SGCC dataset + simulated theft + our testbed as ground truth.

All flags are backed by missing units at the transformer; measure top-20 hit rate

One-page evidence sheet in plain language, plus an approved switch test

Work at transformer level, never break usage down to appliances,

VIABILITY



Technical Viability

- ✓ Model improves with use — every inspection result becomes a new label
- ✓ Retrains on new theft methods without redesign
- ✓ Works on any utility's meter data; no vendor lock-in



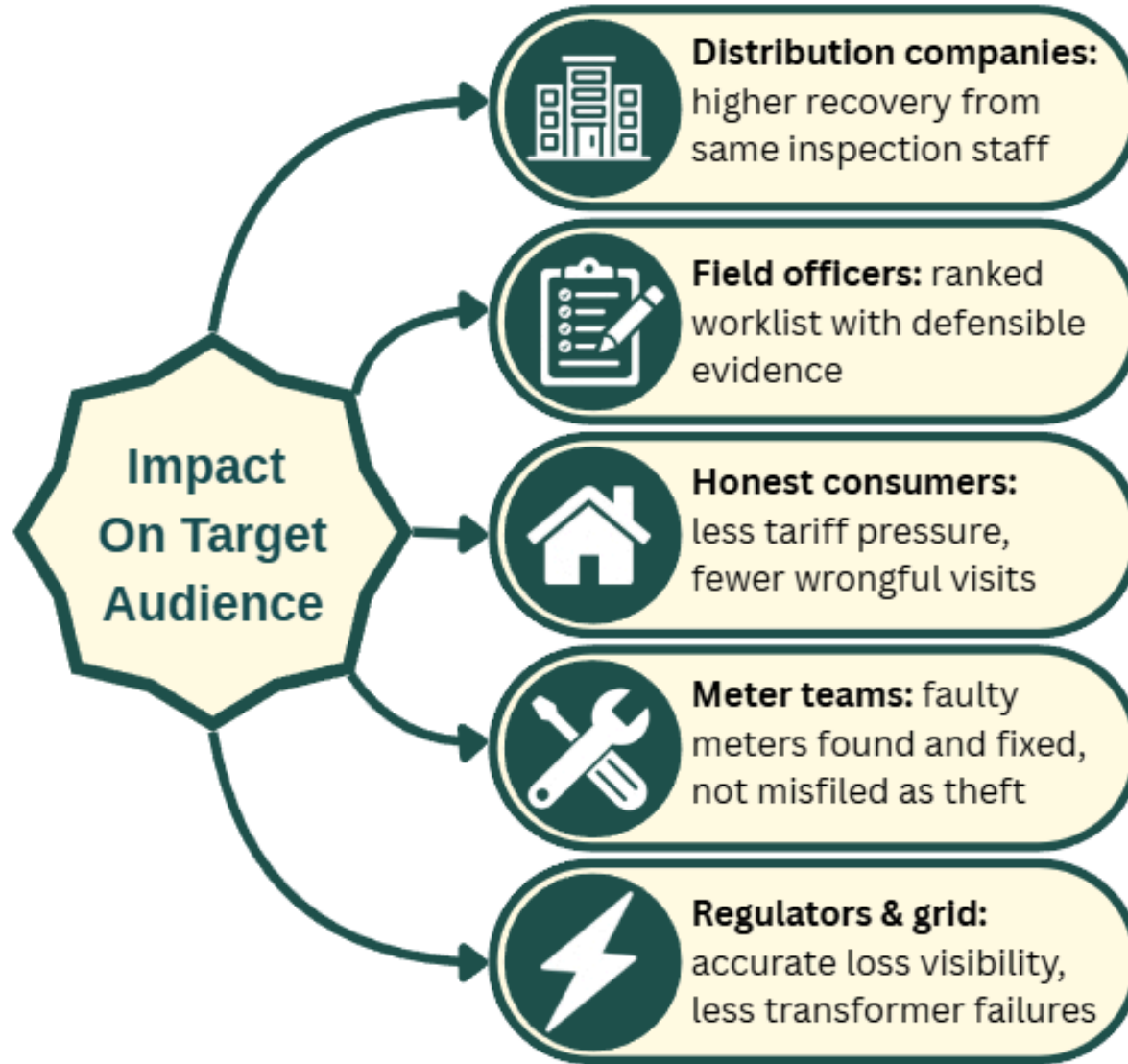
Economic Viability

- ✓ Recovered revenue funds continued operation
- ✓ Scaling costs almost nothing — same software, more meters
- ✓ Aligns with RDSS loss targets that release central funding



Operational Viability

- ✓ Fits existing vigilance workflows — no new roles or process
- ✓ Accuracy improves as inspections accumulate, so trust grows
- ✓ Cause-routing protects honest consumers, keeping the utility's public support



Benifits of The Solution

SOCIAL



Curbing theft eases tariff pressure on honest households



Failing meters get a technician, not an enforcement visit



Fewer transformer overloads means more reliable supply for the whole street

ECONOMIC



Same inspection staff recover more, with no new hires



Runs on meters already installed under RDSS — no new field hardware



Under-reading meters caught early, stopping silent revenue loss

ENVIRONMENTAL



Stolen units are still generated — less theft means less fuel burned



Fewer overloaded transformers burning out and needing replacement



Accurate consumption data improves load planning and reduces peaking generation

- Ministry of Power (PIB) — AI/ML on smart-meter data for theft detection; DT & feeder metering for monthly energy accounting - <https://pib.gov.in/PressReleasePage.aspx?PRID=1906745>
- SGCC Electricity Theft Detection Dataset — 42,372 consumers × 1,036 days, 8.5% theft (primary supervised benchmark) kaggle.com/datasets/bensalem14/sgcc-dataset
- Spatio-Temporal GNN + LSTM for theft detection (2026) — documents the meter-centric limitation our physics layer answers arxiv.org/pdf/2603.20488
- Deep anomaly detection with federated learning — establishes label scarcity and unseen-attack failure of supervised methods ncbi.nlm.nih.gov/pmc/articles/PMC11125161/
- An efficient electricity theft detection based on deep learning — Scientific Reports (baseline comparison) nature.com/articles/s41598-025-93140-z
- PFC / Factly — DISCOM AT&C losses at 15.04% in FY25, billing and collection efficiency trends factly.in/discom-finances-improve-with-lower-atc-losses-and-stronger-collections/
- CSPDCL Director's Report FY 2023–24 — 100% feeder metering completed; rural theft flagged as a standing challenge <https://nsearchives.nseindia.com>
- CSIS — Powering Chhattisgarh: state AT&C losses of 23.28% against a 16% target csis.org/analysis/engaging-indias-electrification-agenda-powering-chhattisgarh